The Data Modeling Process

30-Minute Hands-On Exercise | Week 9 - Section 2A

Learning Objectives: Define a data model and explain why modeling choices matter | List and sequence the data modeling stages in Power BI | Distinguish between OLTP and OLAP systems | Identify what makes a source an ideal starting point for a Power BI model

Instructions: Complete all four parts in 30 minutes. Refer to your Section 2A notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 2A.

1.	A data model describes the __________________________ of your data and how each part relates to the others.
2.	The first stage of the modeling process is to __________________________ the source data by learning its existing shape and which values will be useful.
3.	In relational databases, the relationship between two tables is frequently recorded using a __________________________ .
4.	A database optimized for reporting, structured with star or snowflake schema, and commonly called a Data Warehouse is known as an __________________________ system.
5.	The ideal Power BI model organizes data into a __________________________ schema.
6.	Adding calculations, hierarchies, and date attributes to the model is called the __________________________ stage.

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term.

Term
A. Data Model
B. OLTP System
C. OLAP System
D. Source Model

Description
1. The original shape of data before it has been modeled or transformed
2. A constraint in relational databases that points to a related row in another table
3. A database structured for reporting; less normalized, uses star/snowflake schema
4. The ideal target shape for a Power BI semantic model

E. Destination Model	5. The shape that describes relationships within a dataset and how each part relates
F. Foreign Key	6. Processes many simple, fast transactions; highly normalized; multi-user; emphasizes data integrity

Part 3: Modeling Stages in Order (~8 minutes)

The six data modeling stages are listed below in order. For each stage, write at least two key actions or questions that belong to that stage in the right-hand column.

Order	Modeling Stage	Key Action(s) in this Stage
1	**1. Discover**	*Write 2+ actions here*
2	**2. Bring In (Ingest)**	*Write 2+ actions here*
3	**3. Profile**	*Write 2+ actions here*
4	**4. Transform**	*Write 2+ actions here*
5	**5. Clean**	*Write 2+ actions here*
6	**6. Augment**	*Write 2+ actions here*

Part 4: Short Answer (~9 minutes)

Answer each question in 2-4 sentences in your own words.

Question 1

A junior analyst receives a sales CSV file and immediately starts building visuals. What critical step(s) of the modeling process are they skipping, and why does that matter?

Question 2

Your source data uses the data type 'Float' for a price column. You want report users to see values displayed as currency (e.g., $1,200.00). At which modeling stage would you address this, and why is Power BI better suited for it than a traditional relational database?

Question 3

What is the difference between a 'simple' calculation and an 'advanced' calculation in the Augment stage? Give a concrete example of each.

Question 4

Your company's data lives in a highly normalized OLTP database with 40 tables. A colleague suggests connecting Power BI directly to it without any modeling changes. What two problems could arise, and what would be a better approach?

Bonus / Reflection (if time allows)

Section 2A states: 'Understanding your source data is essential to the modeling process -- and understanding an ideal destination model is just as essential.' In 3-5 sentences, explain what this means in practice. Why is it not enough to understand just one side?
